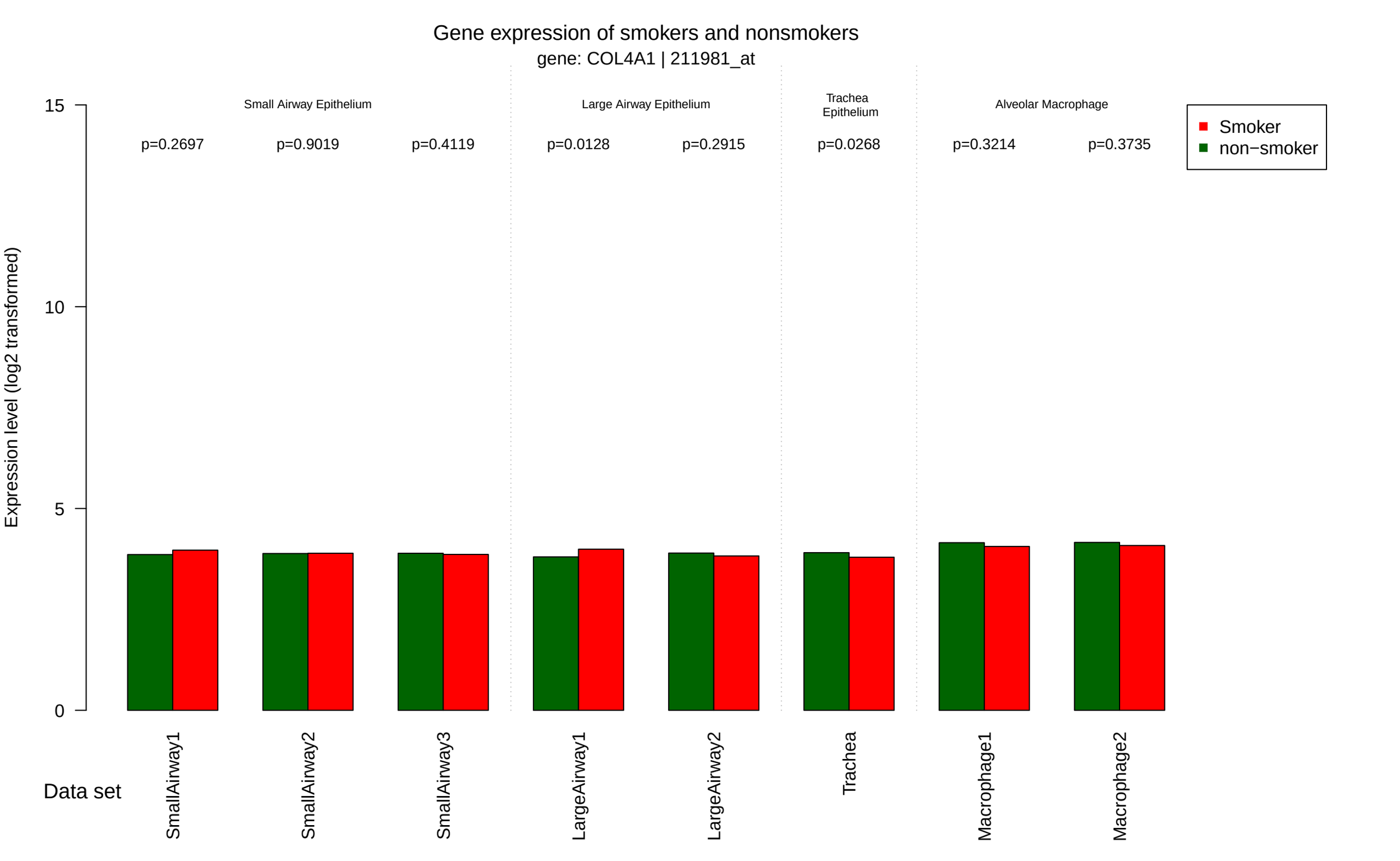
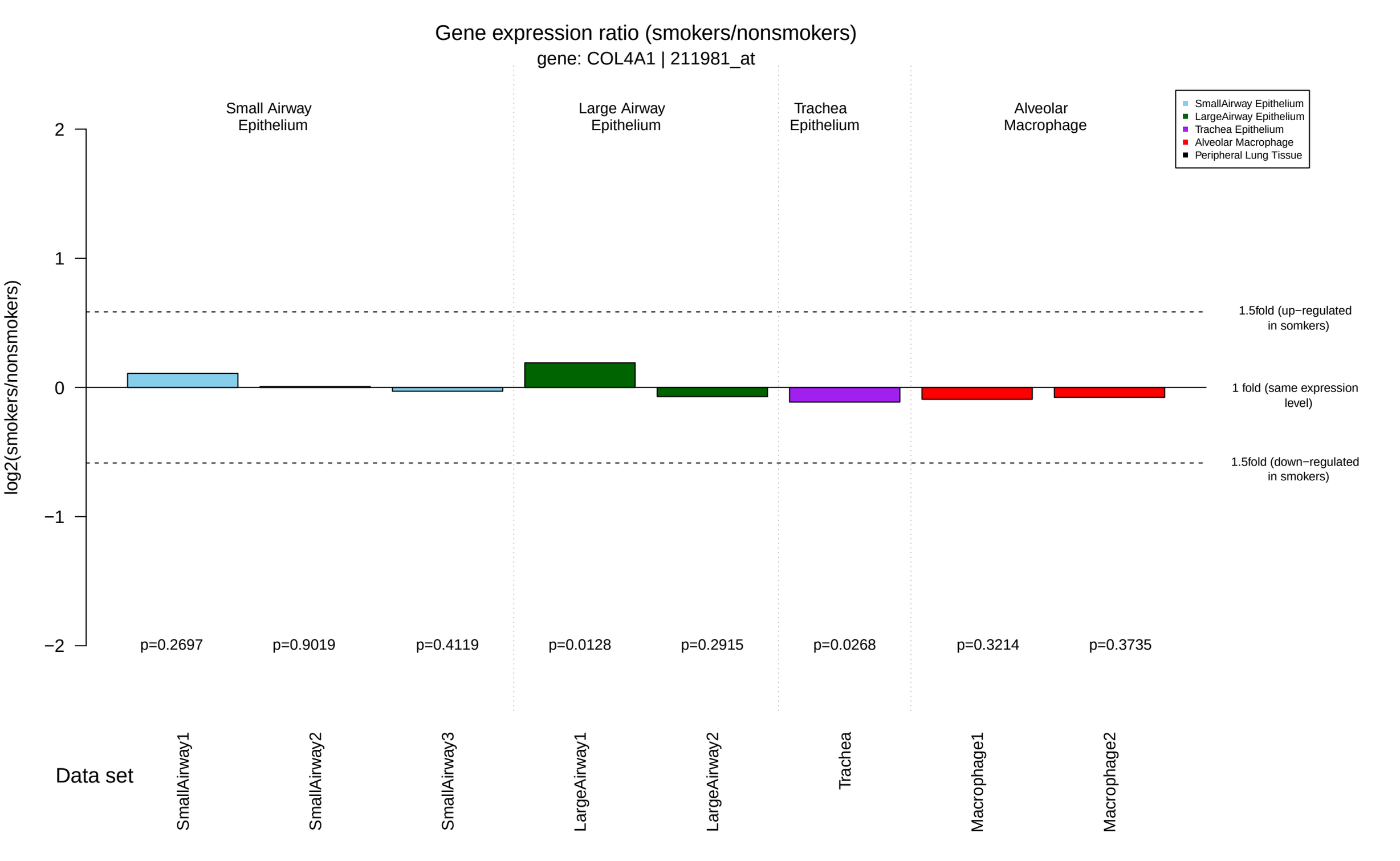
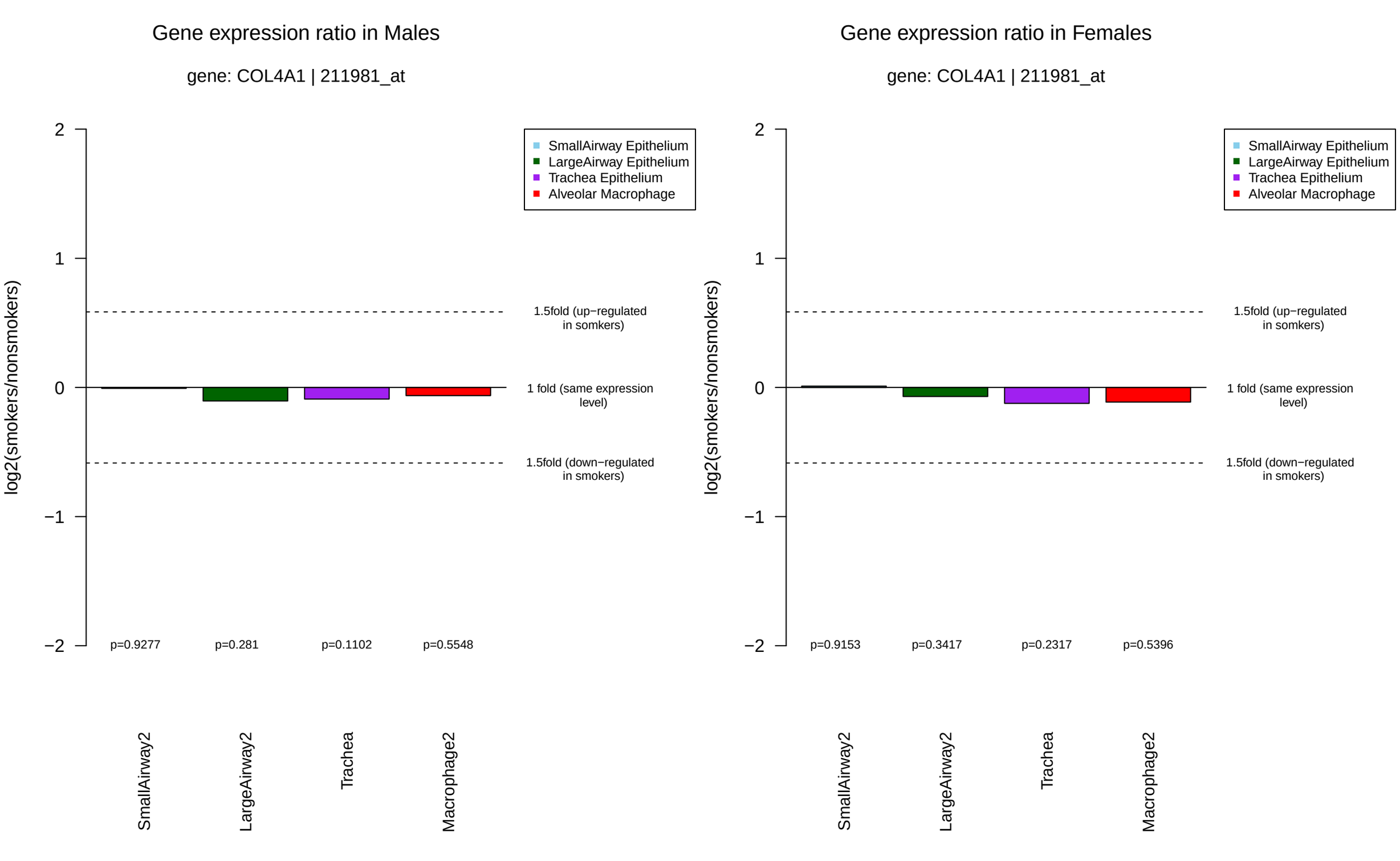
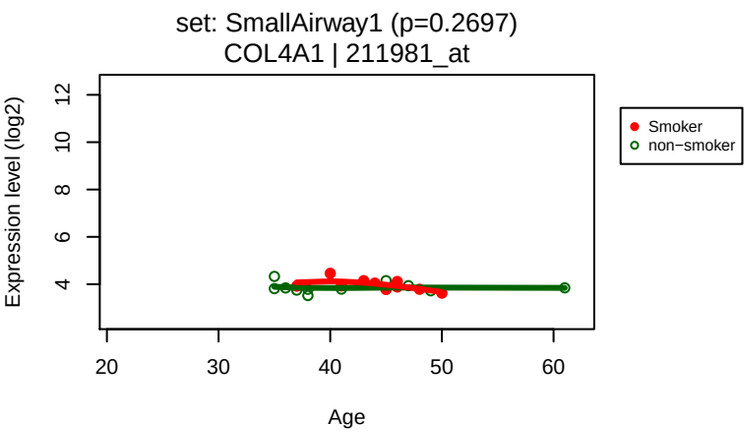
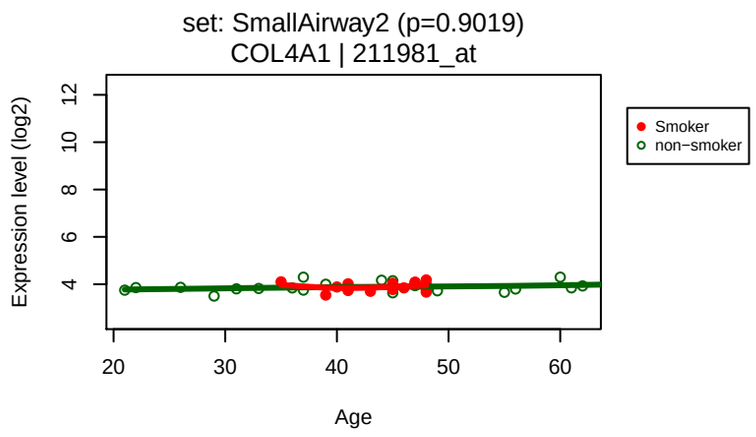
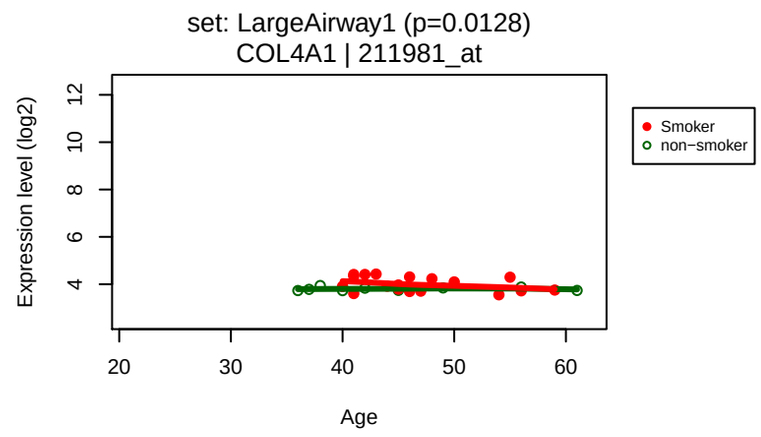

Gene expression of smokers and nonsmokers
gene: COL4A1 | 211981_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2697
p=0.9019
p=0.4119
p=0.0128
p=0.2915
p=0.0268
p=0.3214
p=0.3735
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

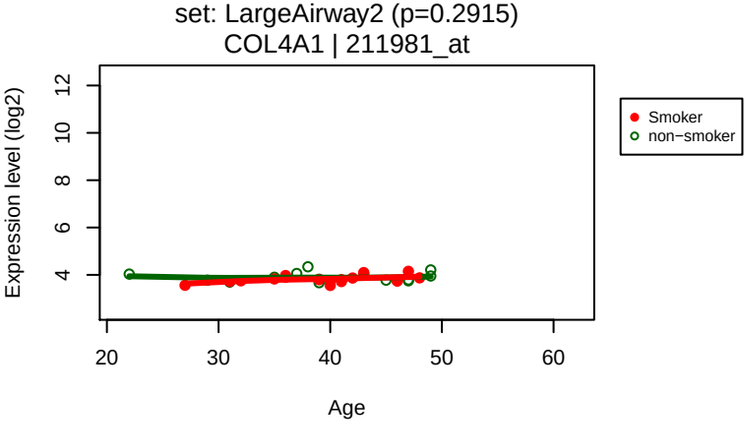

Gene expression ratio (smokers/nonsmokers)
gene: COL4A1 | 211981_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2697
p=0.9019
p=0.4119
p=0.0128
p=0.2915
p=0.0268
p=0.3214
p=0.3735
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

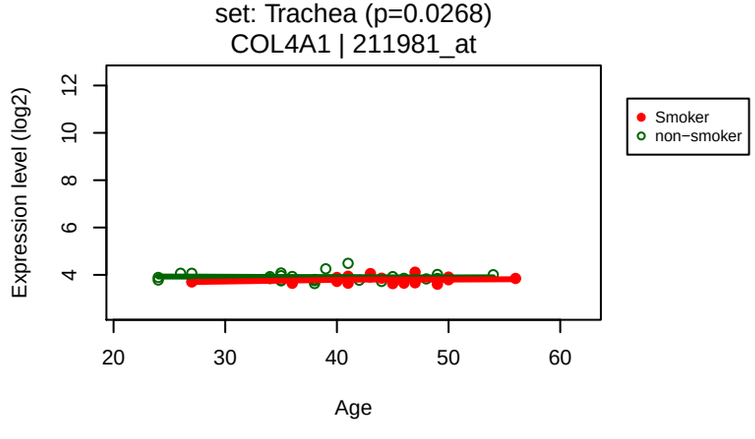

Gene expression ratio in Males
gene: COL4A1 | 211981_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9277
p=0.281
p=0.1102
p=0.5548
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COL4A1 | 211981_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9153
p=0.3417
p=0.2317
p=0.5396
SmallAirway2
LargeAirway2
Trachea
Macrophage2

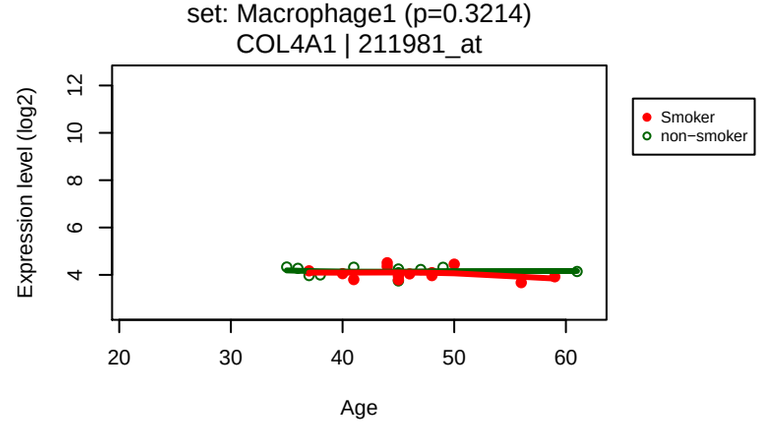

set: SmallAirway1 (p=0.2697)
COL4A1 | 211981_at
Expression level (log2)
Age
Smoker
non−smoker

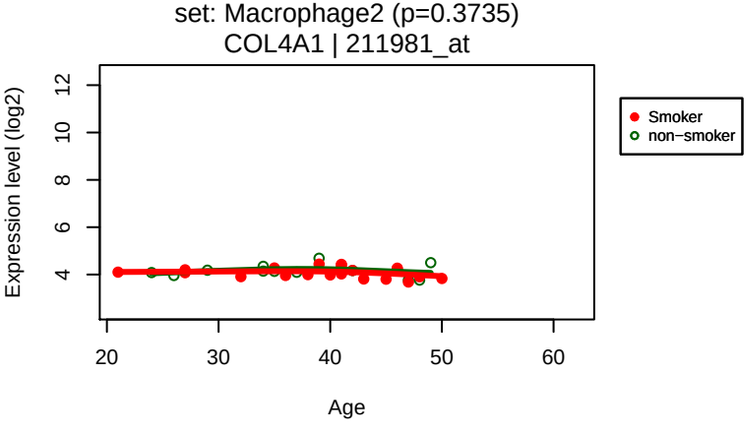

set: SmallAirway2 (p=0.9019)
COL4A1 | 211981_at
Expression level (log2)
Age
Smoker
non−smoker

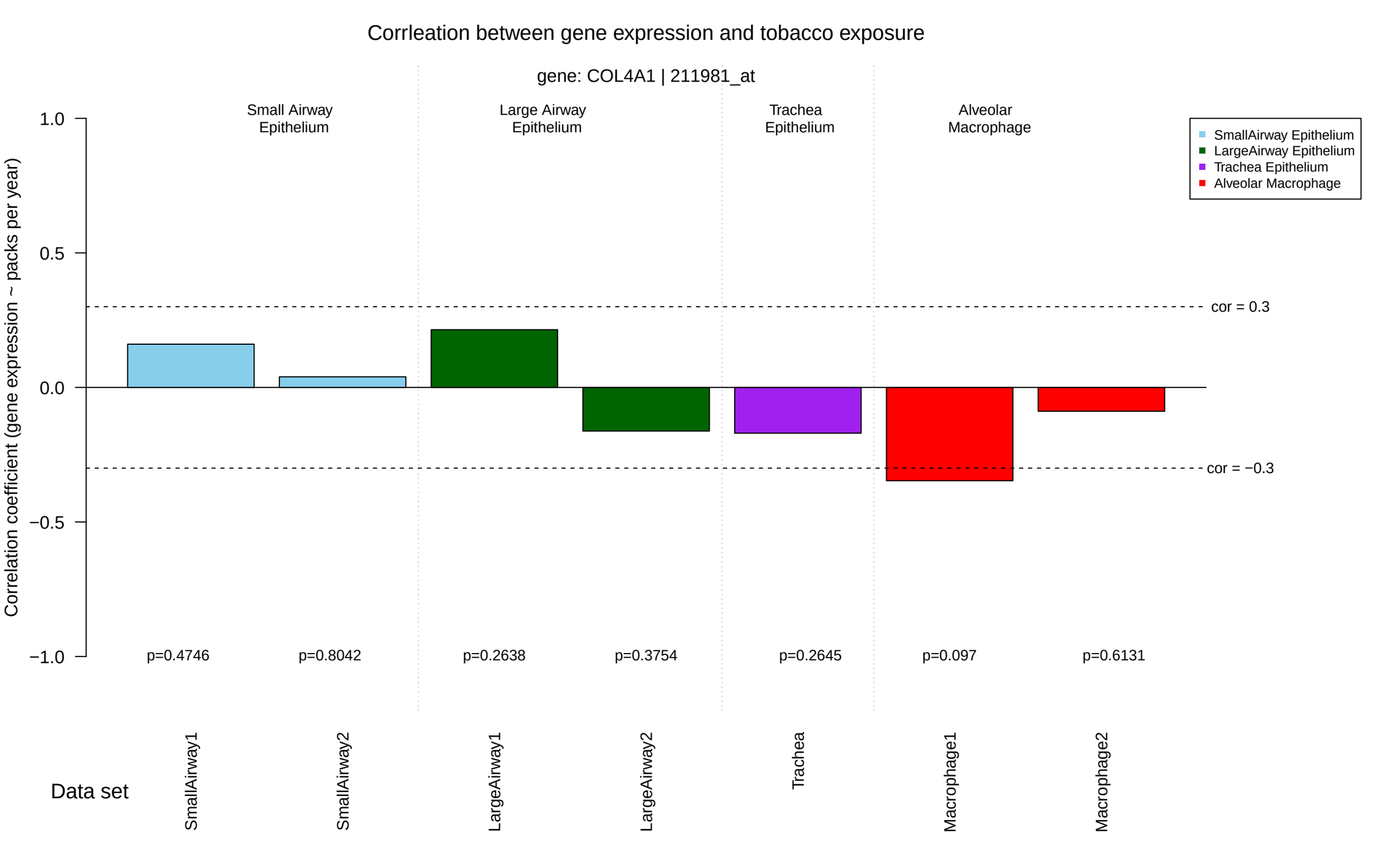

set: LargeAirway1 (p=0.0128)
COL4A1 | 211981_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2915)
COL4A1 | 211981_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0268)
COL4A1 | 211981_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3214)
COL4A1 | 211981_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3735)
COL4A1 | 211981_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COL4A1 | 211981_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4746
p=0.8042
p=0.2638
p=0.3754
p=0.2645
p=0.097
p=0.6131
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2